PERSONAL FINANCE MANAGEMENT SYSTEM - REPORT 1

GENERAL OVERVIEW

BY: ARJOLA METALIAJ, EDERTA KARIQI, ASJA XHELEZI, STILIANOS TROKA, ELDA TOSKA, SINDI TERIHATI, ELISA SPAHO

Personal Finance Management System

TOPIC AND AIM

1. Project Topic

The project focuses on the design and specification of a Personal Finance Management System (PFMS).

The system helps users manage financial activities by tracking expenses, monitoring savings goals, and analyzing financial behavior.

This project represents a more advanced system design stage of a previously implemented Personal Finance Management System developed in the Object-Oriented Programming course. While the previous project focused on implementation, this project focuses on requirements engineering, system analysis, and software design modelling.

2. Aim of the System

The aim of the system is to provide a structured platform for tracking, analyzing, and improving personal financial behavior.

The system allows users to:

- record and categorize expenses
- monitor savings progress
- define financial budgets
- analyze spending patterns
- receive automated financial insights

The system follows a layered architecture separating user interface, business logic, and data management, ensuring modularity and maintainability.

3. User Levels

Standard User
The primary user of the system who manages personal financial data.
Capabilities:
- register and authenticate
- record expenses and income
- categorize financial transactions
- set savings goals
- view financial dashboards and reports

Advanced User
Users who access extended analytical and budgeting features.
Capabilities:
- configure detailed budgets
- analyze spending trends
- receive automated financial insights

System Administrator
Responsible for system maintenance and management.
Responsibilities:
- manage user accounts
- monitor system activity
- maintain system configuration and data integrity

4.1 Functional Requirements

Functional requirements describe what the system shall do.

User Management
- The system shall allow users to register an account.
- The system shall allow users to log in securely.
- The system shall allow users to update profile information.

Expense Management
- The system shall allow users to record expenses.
- The system shall allow categorization of financial transactions.
- The system shall allow editing and deletion of expense records.

Savings and Budget Management
- The system shall allow users to define savings goals.
- The system shall calculate savings progress.
- The system shall allow users to define monthly budgets.
- The system shall notify users when expenses exceed defined budgets.

Financial Analysis
- The system shall analyze financial records.
- The system shall generate spending summaries and reports.
- The system shall provide financial insights based on user activity.

Data Storage
- The system shall store financial records persistently.
- The system shall allow retrieval of historical financial data.

4.2 Non-Functional Requirements

Non-functional requirements describe system quality attributes and operational constraints.

Performance
The system shall respond to user actions within an acceptable response time.

Usability
The interface should be simple and easy to use, allowing users to perform financial tasks with minimal training.

Security
The system shall ensure that financial data is accessible only to authorized users.

Reliability
The system should ensure consistent storage and retrieval of financial records.

Maintainability
The system architecture should support modular design and future extensions.

5. Stakeholders

Stakeholders are individuals or groups who interact with or are affected by the system.

The main stakeholders include:
- system users managing personal finances
- administrators maintaining the system
- developers responsible for system development and maintenance